All solutions to graded problems must be typed in LaTeX. All other problems can be typed or written, but must be completed. Graded problems must be organized in numerical order. All problem sets (both graded and non-graded) must be uploaded to Canvas as a single PDF document. Students can find the grading scale in our Syllabus.

Graded

Exercise 1. Prove that if A is closed in X and B is closed in Y, then $A \times B$ is closed in $X \times Y$.

Exercise 2. Find the closure and interior of the following sets in $\mathbb{R}^2$.

1. $A = \{x \times y : y = 0\}$
2. $B = \{x \times y : x > 0 \text{ and } y > 0\}$
3. $C = \{x \times y : x^2 + y^2 \leq 1 \text{ and } x > 0\}$
4. $D = \{x \times y : x > 0 \text{ and } y = \sin(\frac{1}{x})\}$

Non-Graded

Exercise 3. Prove that if X is a Hausdorff space, then $X \times X$ is a Hausdorff space.

Exercise 4. Show that X is Hausdorff if and only if the set

$$\Delta = \{x \times x : x \in X\}$$

is closed in $X \times X$.

Exercise 5. In the cofinite topology on $\mathbb{R}$, to what point or points does the sequence $\{1/n : n \in \mathbb{Z}_+\}$ converge?